Modeling VR Worlds

Author: Abdelkrim Mebarki
Master's Program in Computer Science – Artificial Intelligence & Applications
Department of Computer Science – Faculty of Mathematics and Computer Science
University of Science and Technology of Oran
November 16, 2025

To **perceive** a virtual world, the user needs entities to be **rendered** on devices. But before rendering can happen, that world must first **exist in a digital form** — as a **coherent structured representation** of all its objects, spaces, and interactions.

What does this mean?

➔ The **virtual world** the user sees, hears, or feels does not exist physically (No external existence).
➔ It must be **created**, **described**, and **encoded** in a form the computer can understand and process.
➔ This internal digital description is what we call the **VR World Representation or VR Modeling**.

To create a Virtual Reality (VR), we must first **model** the virtual world — that is, **design and implement its digital representation** so that it can later be **rendered** and **experienced** through immersive devices.

1. Purpose of Modeling

It defines the **structure**, **behavior**, and **meaning** of the elements composing the virtual world. Without accurate models, rendering (visual, auditory, haptic) cannot produce coherent or immersive experience.

2. Types of Rendering

Once the world is modeled, the VR system can render it through various sensory channels:

- **Visual Rendering**: Transforms 3D models into 2D images displayed to the user.
- **Auditory Rendering**: Spatializes sounds based on object position and user orientation.
- **Haptic Rendering**: provides tactile or force feedback corresponding to object interaction.

VR applications combine all three to achieve **multisensory immersion**.

3. Modeling Constraints

A valid virtual world model has to satisfy 6 **constraints** to ensure realism and coherence. When these constraints are fulfilled, the user perceives the virtual world as **credible** and become **immersed** in it:

1. **Geometric consistency** – Objects have realistic shape, size, and proportions.
2. **Spatial coherence** – Positions, distances, and orientations respect physical logic.
3. **Physical plausibility** – Interactions follow basic physical rules (gravity, collision, resistance).
4. **Temporal continuity** – Smooth and consistent object movements and transformations over time.
5. **Semantic consistency** – Each object's meaning and function are preserved in context.
6. **Computational efficiency** – the model must be optimized for real-time rendering performance.

4. The Visual Model: The Core

The **visual model** is the foundation of any virtual world. The other modalities (such as sound and haptics) are **added on top** to enrich user perception and enhance immersion. The design of the visual model typically involves **four key stages**:

4.1. The Geometric Modeling

This model defines the **shapes and the structure** of objects and spaces using mathematical representations (meshes, surfaces, volumes) to describe the world geometry.

4.2. The Kinematic Modeling

It defines the **movements and the transformations** (translation, rotation, scaling) that affect virtual entities over time. This step introduces **animation** and **user-object motion**.

4.3. The Topological Modeling

This model defines **spatial relationships** between the virtual objects — adjacency, connectivity, inclusion. It ensures spatial continuity (e.g., doors connected to rooms, walls enclosing spaces).

4.4. The Semantic Modeling

Defines **the behavioral and interaction rules**: Collisions, interferences, logical relations, user interactions giving **meaning** to actions. e.g. what happens when the user touches, moves, or manipulates an object.

Available on zenodo – DOI : 10.5281/zenodo.17627209

Modeling Type	Purpose	Impact on Immersion
Geometric	Defines shapes and spaces	Supports visual realism
Kinematic	Controls motion and animation	Increases interactivity
Topological	Structures spatial relationships	Enhances spatial presence
Semantic	Defines interaction logic	Builds meaning and coherence
Auditory/Haptic	Extend sensory realism	Deepen immersion

5. Integration of Multisensory Models

Once the **visual core** is ready, it's time to integrate:

- **Sound models** (spatialized, reactive audio cues)
- **Haptic models** (touch, texture, or force feedback)

These additional layers enhance the **sense of presence** and **interactivity**, the two pillars of immersion.

5.1. 🎧✋ Integration of Sound and Haptic Modalities within the Modeling Stages

The **visual model** is the structural foundation, **sound and haptic modalities** are **integrated progressively in the final design** — mostly during the **semantic stage**, but they rely on the **information built in the previous stages**:

🧱 1 **Geometric Modeling**

➡️ *Base for spatial references*

- Sound sources and haptic zones need **positions and shapes** to exist in the virtual space.
- Example: defining where a loudspeaker or a vibrating object is located.

🟢 *Sound and haptics are not yet active here, but geometry provides the necessary coordinates.*

🔄 2 **Kinematic Modeling**

➡️ *Movement and transformation of sensory sources*

- When objects move, **their associated sounds and haptic feedback** have to follow.
- Example: a drone's buzzing sound moves with the drone's trajectory.

🟡 *Sound/haptic parameters begin to depend on motion data.*

🌐 3 **Topological Modeling**

➡️ *Spatial and adjacency logic for sensory propagation*

- Defining **how sound travels** (through doors, between rooms) and **where haptic effects apply**.
- Example: footsteps' sound changes between rooms; vibration passes through connected surfaces.

🟡 *Topology defines the logic of sensory continuity and propagation.*

🧠 4 **Semantic Modeling**

➡️ *Integration and activation of multisensory interactions*

- This is where **sound and haptic models** are truly **implemented and controlled**, Defining **rules**, **triggers**, and **responses** linking user actions to multisensory outputs
- Example: When the user touches an object → **vibration + impact sound**

🟢 *Full multisensory integration happens here.*

Modeling Stage	Role for Sound & Haptics	Integration Level
Geometric	Defines spatial location of sensory sources	🟢 Basic
Kinematic	Links sensory cues to object movement	🟡 Moderate
Topological	Controls sensory propagation and adjacency	🟡 Moderate
Semantic	Defines interaction rules and sensory feedback	🔴 Full integration

Sound *and* ***haptics*** *are anchored in the* ***geometry****, animated in the* ***kinematics****, propagated in the* ***topology****, and activated in the* ***semantics****.*

6. 🧩 Classification of Objects in Virtual Worlds

1 Rigid and Non-Deformable Solids 🧱

Objects that do **not deform** — Only undergo **rigid transformations** (translation, rotation, scaling).

➔ Represent static or mechanical elements such as **buildings, furniture, or vehicles**.

✅ *Fundamental for geometry, physics, and collisions.*

Approach – 🧮 **Polygonal / Mesh-Based Modeling** + **Rigid-Body Physic** ➔ No vertex deformation.

2 Articulated Solids ⚙️

Systems of **rigid parts connected by joints** (axes of rotation, sliding, or hierarchy).

➔ Used for **skeletal or mechanical structures**: **avatars, robots, doors, or arms**.

✅ *Essential for animation and kinematics.*

Approach – 🦴 **Skeletal Modeling** with **Forward/Inverse kinematics** (FK/IK) + **joint constraints**.

3 Deformable Solids 🍬

Objects **changing shape or volume** according to **physical or procedural deformation laws**.

➔ **Usage:** Common for **organic materials** (skin, muscles, clothes, rubber, vegetation).

✅ *Crucial for realism in soft materials and living beings.*

Approach – 💾 **Soft-Body Physics**, **Finite Element Modeling (FEM)**, **Blendshape Animation**

→ Vertex deformation.

4 Non-Solids (Continuous Media) 💧

Matter without a fixed shape or volume: **fluids, gases, fire, smoke, or particle systems**.

➔ Used to simulate **natural environments**, **weather**, or **dynamic visual effects**.

✅ *Enhanced through GPU-based simulation and shaders.*

Approach – 🎇 **Particle Systems**, **Fluid Dynamics (SPH, Navier–Stokes)**, **Volumetric rendering**.

5 Abstract or Procedural Entities 🌀 *(Modern Extension)*

Not physical, **logical, interactive, data-driven** (e.g., holograms, force fields, UI elements, or AI agents).

➔ Increasingly used in **XR**, **data visualization**, and **interactive interfaces**.

🆕 *Emerging category — bridges modeling and interaction logic.*

Modeling Approach – 🧠 **Procedural modeling**, **event-driven logic**, or **parametric generation**.

Category	Behavior	Examples	Modeling Approach
🧱 Rigid Solids	No deformation	Buildings, tools	Mesh modeling + rigid-body physics
⚙️ Articulated Solids	Rigid parts with joints	Robots, avatars	Skeletal modeling + kinematics
🧬 Deformable Solids	Elastic deformation	Skin, cloth	Soft-body / FEM / blendshapes
💧 Non-Solids	No fixed volume	Water, fire, smoke	Particle / fluid dynamics
🌀 Abstract Entities	Logical / procedural	UI, AI, holograms	Procedural / event-driven

7. 🧱 Formats of Geometric Representation in Virtual Worlds

Back to the geometric model, the core of the virtual world representation: Geometric structures can be represented using **two main approaches**: **Primitive-based models** or **Analytical** models.

1 Primitive-Based Representation 🧱

Built from **basic geometric elements** combined to form complex shapes. This is the most common format used in real-time 3D and VR applications including :

- 🔹 **Vertex-based:** Defined by 3D points (vertices) in space.
- 🔹 **Edge-based:** Defined by connections between vertices (edges).
- 🔹 **Face-based:** Defined by polygonal surfaces (triangles, quads, meshes).

These entities (polygonal and meshes) are suitable for **interactive rendering** and **collision detection**.

2 Analytical Representation 📐

Described using **mathematical equations** instead of discrete primitives focusing on **precision** and **continuous surfaces**. This includes:

- 🔸 **Parametric models –** Surfaces defined by **mathematical functions** (e.g., Bézier, B-splines, NURBS).
  - 🧩 Used for CAD, organic shapes, and smooth curves.
- 🔸 **Non-parametric models –** Defined **implicitly or explicitly** by equations without parameters.
  - 🧱 **Example:** Implicit: F(x,y,z)=0 | Explicit: z=f(x,y)
  - 🌋 Used for volumetric forms, terrains, or implicit surfaces (metaballs).

Equation-based and procedural models suit **scientific visualization**, **simulation**, or **precise design**.

Representation	Basis	Typical Use	Strengths	Limitations
🧱 Primitive-based	Vertices, edges, faces	Games, VR, real-time scenes	Fast, intuitive, GPU-friendly	Less precise, discrete
📈 Analytical	Mathematical equations	CAD, simulation	Smooth, exact	Harder to render in real time